CUSTOMER CHURN ANALYSIS AND RETENTION STRATEGY

A Data-Driven Business Intelligence Project

1. Project Overview

Customer churn occurs when customers discontinue their services or subscriptions. Understanding churn is important for businesses because retaining existing customers supports long-term growth and customer loyalty.

This project analyses customer data using Python and statistical techniques to identify patterns associated with customer churn and develop data-driven retention recommendations.

2. Dataset and Methodology

The analysis used a dataset containing **500 customer records and 9 attributes**, including tenure, monthly charges, total charges, contract type, payment method, paperless billing, and churn status.

The project followed three major stages:

- **Data Cleaning:** Checked missing values, duplicate records, duplicate customer IDs, and data consistency. No major data quality issues were identified.
- **Exploratory Data Analysis:** Examined customer distribution, churn rates, tenure, charges, contract types, and payment methods.
- **Statistical Analysis:** Applied Welch's independent samples t-tests and a Chi-square test of independence to evaluate relationships between customer characteristics and churn.

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy, and Jupyter Notebook.

3. Key Business Findings

Key Performance Indicator	Result
Total Customers	500
Retained Customers	447 (89.4%)
Churned Customers	53 (10.6%)
Average Customer Tenure	36.53 months

Major findings:

- **Customer Tenure:** Customers who churned had an average tenure of 6 months, compared with 40.15 months among retained customers.
- **Monthly Charges:** Churned customers had higher average monthly charges (129.77) than retained customers (111.72).
- **Contract Type:** Month-to-month customers had the highest observed churn rate at 20.59%, compared with 4.30% for one-year contracts and 6.94% for two-year contracts.
- **Payment Method:** Credit Card customers had the highest observed churn rate among payment methods, at 13.48%.

4. Statistical Evidence

At a 5% significance level, all three statistical tests identified significant differences or associations:

- Tenure and churn: p-value = 8.98×10^{-130}.
- Monthly charges and churn: p-value = 0.0101.
- Contract type and churn: p-value = 9.59×10^{-7}.

These results indicate statistically significant relationships in the analysed dataset. They do not establish causation.

5. Business Recommendations

Based on the findings, the following actions are recommended:

1. **Improve Customer Onboarding:** Strengthen early customer engagement, particularly during the first few months.
2. **Review Pricing Strategies:** Evaluate monthly pricing and communicate service value clearly.
3. **Encourage Long-Term Contracts:** Consider suitable incentives for customers who choose longer subscription plans.
4. **Monitor Customer Retention:** Track churn rate, retention rate, and customer tenure regularly.
5. **Investigate Payment Preferences:** Examine whether payment experience or customer characteristics help explain differences across payment methods.

6. Conclusion

The analysis identified a churn rate of **10.6%** among 500 customers. Customer tenure, monthly charges, and contract type showed statistically significant relationships with churn.

The findings provide a foundation for targeted customer retention strategies and demonstrate how Python-based data analytics can support business decision-making.

Project Title: Customer Churn Analysis and Retention Strategy
Project Type: Data Analytics and Business Intelligence